

DATABASE MANAGEMENT SYSTEMS (CS301)

SQL Joins

Learning Objectives

Understand SQL joins, their types, syntax, examples, and applications in combining data from multiple tables.

Introduction to Joins

A JOIN clause is used to combine rows from two or more tables based on a related column between them.

Need for Joins

In relational databases, data is often stored in multiple tables. Joins allow us to retrieve related information from these tables in a single query.

Sample Tables

Student(StudentID, Name, DeptID)
Department(DeptID, DeptName)

INNER JOIN

Returns only matching records from both tables.

```
SELECT Name, DeptName FROM Student INNER JOIN Department ON Student.DeptID =  
Department.DeptID;
```

LEFT JOIN

Returns all records from the left table and matching records from the right table.

```
SELECT Name, DeptName FROM Student LEFT JOIN Department ON Student.DeptID =  
Department.DeptID;
```

RIGHT JOIN

Returns all records from the right table and matching records from the left table.

```
SELECT Name, DeptName FROM Student RIGHT JOIN Department ON Student.DeptID =  
Department.DeptID;
```

FULL OUTER JOIN

Returns all matching and non-matching records from both tables.

```
SELECT Name, DeptName FROM Student FULL OUTER JOIN Department ON Student.DeptID =  
Department.DeptID;
```

SELF JOIN

A table is joined with itself to compare rows within the same table.

Used in employee-manager relationships and hierarchical data.

Join Comparison

INNER JOIN → Matching rows only
LEFT JOIN → All left table rows
RIGHT JOIN → All right table rows
FULL JOIN → All rows from both tables
SELF JOIN → Table joined with itself

Applications of Joins

- Student-Course Management
- Banking Systems
- E-Commerce Applications
- Employee Management Systems

Exam Important Questions

1. What is a JOIN?
2. Explain INNER JOIN with example.
3. Differentiate LEFT and RIGHT JOIN.
4. Explain FULL OUTER JOIN.
5. What is SELF JOIN?

Summary

SQL Joins are essential for retrieving related data from multiple tables. Understanding different join types is important for database design, querying, and examinations.